Appendix C: Matlab Codes

code_1_Leaky_Integrate_and_fire.m:

```
El = -70e-3;
Rm = 50e6;
Gl = 1/Rm;
Cm = 250e-12;
Vth = -50e-3;
Vreset = -65e-3;
tmax = 2;
dt = 0.001;
t = dt:dt:tmax;
Vm = zeros(size(t));
Vm(1) = El;
I0 = 425e-12;
Iapp = zeros(size(t));
Iapp(0.5/dt:1.5/dt) = I0;
spikes = zeros(size(t));
for i = 2:length(t)
   Vm(i) = Vm(i-1) + dt*(Gl*(El-Vm(i-1))+Iapp(i))/Cm;
   if(Vm(i)>Vth)
       Vm(i) = Vreset;
       spikes(i) = 1;
   end
end
clf
figure(1)
subplot(3,1,1)
plot(t,Iapp);
ylabel("Iapp(A)")
subplot(3,1,2)
plot(t,Vm);
ylabel("Vm(V)")
subplot(3,1,3)
plot(t,spikes);
ylabel("spikes")
```

code_2_firing_rate.m:

```
El = -70e-3;
Rm = 50e6;
Gl = 1/Rm;
Cm = 250e-12;
Vth = -50e-3;
Vreset = -65e-3;
Imax = 900e-12;
dI = 0.1e-12;
Iapp = (dI:dI:Imax);
fire_rate = zeros(size(Iapp));
for i = 1:length(Iapp)
   if(El+Iapp(i)/Gl>Vth)
       fire_rate(i)=1/(Cm*Gl*log((El+Iapp(i)/Gl-Vreset)/(El+Iapp(i)/Gl-Vth)));
   end
end
clf
figure(1)
plot(Iapp,fire_rate);
xlabel("applied curret (A)")
ylabel("firing rate (Hz)")
```

code_3_refractory_period:

```
tmax = 2;
dt = 0.0001;
t = 0:dt:tmax;
El = -70e-3;
Rm = 100e6;
Cm = 0.1e-9;
Iapp = 100:600;
Iapp = Iapp*1e-12;
methods = 1:3;
Vpeak = 50e-3;
```

```
Vreset = -65e-3;
for method = 1:length(methods)
   fire_rate = zeros(size(Iapp));
   mean_Vm = zeros(size(Iapp));
   if(method == 1)
       Tref = 2.5e-3/dt;
       Vth = -50e-3;
       for i = 1:length(Iapp)
           fire_times = 0;
           Vm = zeros(size(t));
           Vm(1) = El;
           Tlast_spike = -2*Tref;
           for j = 2:length(t)
               if(j <= Tlast_spike + Tref)
                   Vm(j) = Vreset;
               else
                   Vm(j) = Vm(j-1) + dt*((El - Vm(j-1))/Rm + Iapp(i))/Cm;
                   if(Vm(j) > Vth)
                       Vm(j-1) = Vpeak;
                       Vm(j) = Vreset;
                       Tlast_spike = j;
                       fire_times = fire_times+1;
                   end
               end
           end
           fire_rate(i) = fire_times/tmax;
           mean_Vm(i) = mean(Vm);
           if(i == 120 || i == 500)
               figure(2);
               hold on;
               subplot(2,2,1);
               plot(t(1:600),Vm(1:600));
           end
       end
       figure(1);
       hold on;
       subplot(2,2,1);
       plot(Iapp,fire_rate,'blue');
       xlabel('Iapp');
       ylabel('fire rate');
       hold on
       subplot(2,2,2);
       plot(Iapp,mean_Vm,'blue');
       xlabel('Iapp');
       ylabel('mean Vm');
       hold on
       subplot(2,2,3);
       plot(fire_rate,mean_Vm,'blue');
       xlabel('fire rate');
       ylabel('mean Vm');
   end
   if (method == 2)
       Ek = -80e-3;
       Vthresh = zeros(size(t));
       Vthmax = 200e-3;
       Taoth = 1e-3;
       Taogref = 0.2e-3;
       for i = 1:length(Iapp)
           Vthresh(1) = -50e-3;
           fire_times = 0;
           Vm = zeros(size(t));
           Vm(1) = El;
           Gref = zeros(size(t));
           Gref(1) = 0;
           dG = 2e-6;
           for j = 2:length(t)
               Gref(j) = Gref(j-1) - dt*Gref(j-1)/Taogref;
               Vthresh(j) = Vthresh(j-1) + dt*(Vthresh(1)-Vthresh(j-1))/Taoth;
               Vm(j) = Vm(j-1) + dt*((El-Vm(j-1))/Rm + Iapp(i))/Cm +
dt*Gref(j-1)*(Ek-Vm(j-1))/Cm;
               if(Vm(j)>=Vthresh(j))
```

```
                Vm(j-1) = Vpeak;
                Vm(j) = Vreset;
                Vthresh(j) = Vthmax;
                fire_times = fire_times + 1;
                Gref(j) = Gref(j) + dG;
            end
        end
        fire_rate(i) = fire_times/tmax;
        mean_Vm(i) = mean(Vm);
        if(i == 120 || i == 500)
            figure(2);
            hold on;
            subplot(2,2,2);
            plot(t(1:600),Vm(1:600));
        end
    end
    figure(1);
    hold on;
    subplot(2,2,1);
    plot(Iapp,fire_rate,'green');
    xlabel('Iapp');
    ylabel('fire rate');
    subplot(2,2,2);
    plot(Iapp,mean_Vm,'green');
    xlabel('Iapp');
    ylabel('mean Vm');
    subplot(2,2,3);
    plot(fire_rate,mean_Vm,'green');
    xlabel('fire rate');
    ylabel('mean Vm');
end
if (method == 3)
    Vthresh = zeros(size(t));
    Vthmax = 200e-3;
    Taoth = 1e-3;
    for i = 1:length(Iapp)
        Vthresh(1) = -50e-3;
        fire_times = 0;
        Vm = zeros(size(t));
        Vm(1) = El;
        for j = 2:length(t)
            Vthresh(j) = Vthresh(j-1) + dt*(Vthresh(1)-Vthresh(j-1))/Taoth;
            Vm(j) = Vm(j-1) + dt*((El-Vm(j-1))/Rm + Iapp(i))/Cm;
            if(Vm(j)>=Vthresh(j))
                Vm(j-1) = Vpeak;
                Vm(j) = Vreset;
                Vthresh(j) = Vthmax;
                fire_times = fire_times + 1;
            end
        end
        fire_rate(i) = fire_times/tmax;
        mean_Vm(i) = mean(Vm);
        if(i == 120 || i == 500)
            figure(2);
            hold on;
            subplot(2,2,3);
            plot(t(1:600),Vm(1:600));
        end
    end
    figure(1);
    hold on;
    subplot(2,2,1);
    plot(Iapp,fire_rate,'red');
    xlabel('Iapp');
    ylabel('fire rate');
    subplot(2,2,2);
    plot(Iapp,mean_Vm,'red');
    xlabel('Iapp');
    ylabel('mean Vm');
    subplot(2,2,3);
    plot(fire_rate,mean_Vm,'red');
```

```
        xlabel('fire rate');
        ylabel('mean Vm');
    end
end
```

code_4_Exponential_Leaky_Intergrate_and_Fire.m:

```
El = -70e-3;
Cm = 100e-12;
Gl = 10e-9;
Vth = -50e-3;
Vreset = -80e-3;
dth = 5e-3;
Vmax = 50e-3;
Iapp = 210e-12;
tmax = 0.15;
dt = 0.001;
t = dt:dt:tmax;
Vm = zeros(size(t));
Vm(1) = El;
for i = 2:length(t)
   Vm(i) = Vm(i-1)+dt*(Gl*(El-Vm(i-1)+dth*exp((Vm(i-1)-Vth)/dth))+Iapp)/Cm;
   if (Vm(i) > Vmax)
       Vm(i-1) = Vmax;
       Vm(i) = Vreset;
   end
end
clf
figure(1)
plot(t,Vm);
xlabel("Times (s)");
ylabel("Vm (V)");
```

code_5_AELIF.m:

```
clf
El = -75e-3;
Vth = -50e-3;
Vreset = -80e-3;
Vmax = 50e-3;
dth = 2e-3;
Gl = 10e-9;
Cm = 100e-12;
a = 2e-9;
b = 20e-12;
Taosra = 200e-3;
EK = -80e-3;
dGsra = 1e-9;
tmax = 1;
dt = 0.0001;
Imax = 400e-12;
dI = 4e-12;
Iapp = dI:dI:Imax;
fire_rate = zeros(size(Iapp));
Vmean = zeros(size(Iapp));
t = dt:dt:tmax;
Vm = zeros(size(t));
Isra = zeros(size(t));
Gsra = zeros(size(t));
for i = 1:length(Iapp)
   Vm(1) = El;
   Isra(1) = 0;
   Gsra(1) = 0;
   spikes = zeros(size(t));
   for j = 2:length(t)
       Vm(j) =
Vm(j-1)+dt*(Gl*(El-Vm(j-1)+dth*exp((Vm(j-1)-Vth)/dth))-Isra(j-1)+Iapp(i)+Gsra(j-1)*(EK-Vm(j-1)
))/Cm;
       Gsra(j) = Gsra(j-1)+dt*-Gsra(j-1)/Taosra;
       Isra(j) = Isra(j-1)+dt*(a*(Vm(j-1)-El)-Isra(j-1))/Taosra;
       if(Vm(j)>Vmax)
           Vm(j-1) = Vmax;
```

```
            Vm(j) = Vreset;
            Isra(j) = Isra(j)+b;
            Gsra(j) = Gsra(j)+dGsra;
            spikes(j) = 1;
        end
    end
    if(i==80)
        figure(1)
        subplot(2,2,1)
        plot(t,Vm);
        xlabel("time (s)");
        ylabel("Vm (V)");
    end
    fire_rate(i) = sum(spikes)/tmax;
    Vmean(i) = mean(Vm);
end
figure(1)
subplot(2,2,2)
plot(Iapp,fire_rate);
xlabel("Iapp (A)");
ylabel("Fire rate (Hz)");
subplot(2,2,3);
plot(Iapp,Vmean);
xlabel("Iapp (A)");
ylabel("Mean Valtage (V)");
subplot(2,2,4);
plot(fire_rate,Vmean);
xlabel("Fire rate (Hz)");
ylabel("Mean voltage (V)");
```

code_6_AELIF_with_noise.m:

```
El = -70e-3;
Vth = -50e-3;
Vreset = -80e-3;
dth = 2e-3;
Vmax = 50e-3;
Gl = 10e-9;
Cm = 100e-12;
a = 2e-9;
b = 1e-9;
Taosra = 150e-3;
dt = 0.01e-3;
tmax = 100;
t = dt:dt:tmax;
sigma = 20e-12;
SD = sigma/sqrt(dt);
Iapp = randn(size(t))*SD + 0.2e-12;
Vm = zeros(size(t));
Isra = zeros(size(t));
Vm(1) = El;
spikes = zeros(size(t));
for i = 2:length(t)
    Vm(i) = Vm(i-1) + dt*(Gl*(El-Vm(i-1)+dth*exp((Vm(i-1)-Vth)/dth))-Isra(i-1)+Iapp(i-1))/Cm;
    Isra(i) = Isra(i-1) + dt*(a*(Vm(i-1)-El)-Isra(i-1))/Taosra;
    if(Vm(i)>Vmax)
        Vm(i) = Vreset;
        Vm(i-1) = Vmax;
        Isra(i) = Isra(i) + b;
        spikes(i) = 1;
    end
end
clf
figure(1)
subplot(1,2,1)
spiketimes = dt*find(spikes(:));
isis = diff(spiketimes);
h = histogram(isis,0:25*dt:0.5);
cv = std(isis,1)/mean(isis);
title(strcat('CV=', num2str(cv) ));
bin = 10:1000;
```

```
fano = zeros(size(bin));
for j = 1:length(bin)
   scale = bin(j)*1e-3/dt;
   spikeinwindow = zeros(1,ceil(length(t)/scale));
   for i = 1:length(spiketimes)

spikeinwindow(ceil(spiketimes(i)/dt/scale))=spikeinwindow(ceil(spiketimes(i)/dt/scale))+1;
   end
   fano(j) = std(spikeinwindow,1)^2/mean(spikeinwindow);
end
subplot(1,2,2)
plot(bin,fano);
title(strcat("Fano Factor"))
figure(2)
plot(t,Vm);
xlim([0.5,2.5]);
xlabel("Time (s)");
ylabel("Vm (V)");
```

code_7_Hodgkin_Huxley_Model.m:

```
dt = 0.0001e-3;
tmax = 0.45;
t = dt:dt:tmax;
V = zeros(size(t));
m = zeros(size(t));
h = zeros(size(t));
n = zeros(size(t));
Ibase = 0.6e-9;
Iapp = Ibase*ones(size(t));
npulse = 10;
pduration = 5e-3;
delays = 20e-3*ones(1,10);
Ic = 0;
IC = Ibase*ones(size(dt:dt:(npulse*pduration+sum(delays))));
k=0;
for i = 1:length(delays)
   k = round(k+(delays(i)+pduration)/dt);
   IC(round(k-pduration/dt):k) = Ic;
end
tstart = 100e-3;
tduration = 100e-3;
k = round(tstart/dt);
Iapp(k:(k+length(IC)-1)) = IC(1:length(IC));
alpha_m = zeros(size(t));
alpha_h = zeros(size(t));
alpha_n = zeros(size(t));
beta_m = zeros(size(t));
beta_h = zeros(size(t));
beta_n = zeros(size(t));
Gl = 30e-9;
Cm = 100e-12;
GMNa = 12e-6;
GMK = 3.6e-6;
ENa = 45e-3;
EK = -82e-3;
El = -60e-3;
V(1) = -65e-3;
m(1) = 0.05;
h(1) = 0.5;
n(1) = 0.35;
for i = 2:length(t)
   V(i) = V(i-1) +
dt*(Gl*(El-V(i-1))+GMNa*m(i-1)^3*h(i-1)*(ENa-V(i-1))+GMK*n(i-1)^4*(EK-V(i-1))+Iapp(i-1))/Cm;
   if(V(i)==-0.045)
       alpha_m(i) = 1000;
   else
       alpha_m(i) = 1e5*(-V(i)-0.045)/(exp(100*(-V(i)-0.045))-1);
   end
   beta_m(i) = 4e3*exp((-V(i)-0.070)/0.018);
   alpha_h(i) = 70*exp(50*(-V(i)-0.070));
```

```
   beta_h(i) = 1e3/(1+exp(100*(-V(i)-0.040)));
   if(V(i)==-0.060)
       alpha_n(i) = 100;
   else
       alpha_n(i) = 1e4*(-V(i)-0.060)/(exp(100*(-V(i)-0.060))-1);
   end
   beta_n(i) = 125*exp((-V(i)-0.070)/0.080);
   SS_m = alpha_m(i)/(alpha_m(i)+beta_m(i));
   SS_h = alpha_h(i)/(alpha_h(i)+beta_h(i));
   SS_n = alpha_n(i)/(alpha_n(i)+beta_n(i));
   tao_m = 1/(alpha_m(i)+beta_m(i));
   tao_h = 1/(alpha_h(i)+beta_h(i));
   tao_n = 1/(alpha_n(i)+beta_n(i));
   m(i) = m(i-1) + dt*(SS_m-m(i-1))/tao_m;
   h(i) = h(i-1) + dt*(SS_h-h(i-1))/tao_h;
   n(i) = n(i-1) + dt*(SS_n-n(i-1))/tao_n;
end
clf
figure(1);
subplot(2,1,1);
plot(t,Iapp);
ylabel('Iapp');
subplot(2,1,2);
plot(t,V,'k');
ylabel('V');
figure(2)
hold on;
plot(t,m,'r');
plot(t,h,'g');
plot(t,n,'b');
```

code_8_Connor_Stevens_Model:

```
GL = 30e-9;
GNa = 12e-6;
GK = 2e-6;
GA = 4.77e-6;
ENa = 55e-3;
EK = -72e-3;
EA = -75e-3;
EL = -17e-3;
Cm = 100e-12;
Tmax = 2;
dt = 0.00001; %% Time step needs to be small!!!! If it's not small enough, it can cause the
gating variable to grow beyond the range of 0~1!
t = dt:dt:Tmax;
Ib = (0:100:10000)*1e-12;
Iapp = zeros(length(Ib),length(t));
for i=1:length(Ib)
   Iapp(i,round(0.5/dt):round(1.5/dt)) = Ib(i);
end
Vm = zeros(size(Iapp));
m = zeros(size(Iapp));
h = zeros(size(Iapp));
n = zeros(size(Iapp));
a = zeros(size(Iapp));
b = zeros(size(Iapp));
firerates = zeros(size(Ib));
firecounts = zeros(size(Ib));
spikes = zeros(size(Iapp));
alpha_m = zeros(size(Iapp));
alpha_h = zeros(size(Iapp));
alpha_n = zeros(size(Iapp));
beta_m = zeros(size(Iapp));
beta_h = zeros(size(Iapp));
beta_n = zeros(size(Iapp));
SS_m = zeros(size(Iapp));
SS_h = zeros(size(Iapp));
SS_n = zeros(size(Iapp));
SS_a = zeros(size(Iapp));
SS_b = zeros(size(Iapp));
```

```
Tao_m = zeros(size(Iapp));
Tao_h = zeros(size(Iapp));
Tao_n = zeros(size(Iapp));
Tao_a = zeros(size(Iapp));
Tao_b = zeros(size(Iapp));
Vm(:,1) = -65e-3;
m(:,1) = 0;
h(:,1) = 0;
n(:,1) = 0;
a(:,1) = 0;
b(:,1) = 0;
for i = 1:length(Ib)
   for j = 2:length(t)
       Vm(i,j) =
Vm(i,j-1)+dt*(GL*(EL-Vm(i,j-1))+GNa*m(i,j-1)^3*h(i,j-1)*(ENa-Vm(i,j-1))+GK*n(i,j-1)^4*(EK-Vm(i
,j-1))+GA*a(i,j-1)^3*b(i,j-1)*(EA-Vm(i,j-1))+Iapp(i,j-1))/Cm;
       if(j>=2&&Vm(i,j)>0&&Vm(i,j-1)>Vm(i,j)&&Vm(i,j-1)>Vm(i,j-2))
           spikes(i,j) = 1;
           firecounts(i) = firecounts(i)+1;
       end
       if(Vm(i,j)==-0.0297)
           alpha_m(i,j) = 3.8e3;
       else
           alpha_m(i,j) = 3.8e5*(Vm(i,j)+0.0297)/(1-exp(-100*(Vm(i,j)+0.0297)));
       end
       beta_m(i,j) = 1.52e4*exp(-55.6*(Vm(i,j)+0.0547));
       alpha_h(i,j) = 266*exp(-50*(Vm(i,j)+0.048));
       beta_h(i,j) = 3800/(1+exp(-100*(Vm(i,j)+0.018)));
       if(Vm(i,j)==-0.0457)
           alpha_n(i,j) = 200;
       else
           alpha_n(i,j) = 2e4*(Vm(i,j)+0.0457)/(1-exp(-100*(Vm(i,j)+0.0457)));
       end
       beta_n(i,j) = 250*exp(-12.5*(Vm(i,j)+0.0557));
       SS_m(i,j) = alpha_m(i,j)/(alpha_m(i,j)+beta_m(i,j));
       SS_h(i,j) = alpha_h(i,j)/(alpha_h(i,j)+beta_h(i,j));
       SS_n(i,j) = alpha_n(i,j)/(alpha_n(i,j)+beta_n(i,j));
       SS_a(i,j) = (0.0761*exp(31.4*(Vm(i,j)+0.09422))/(1+exp(34.6*(Vm(i,j)+0.00117))))^(1/3);
       SS_b(i,j) = (1/(1+exp(68.8*(Vm(i,j)+0.0533))))^4;
       Tao_m(i,j) = 1/(alpha_m(i,j)+beta_m(i,j));
       Tao_h(i,j) = 1/(alpha_h(i,j)+beta_h(i,j));
       Tao_n(i,j) = 1/(alpha_n(i,j)+beta_n(i,j));
       Tao_a(i,j) = 3.632e-4+1.158e-3/(1+exp(49.7*(Vm(i,j)+0.05596)));
       Tao_b(i,j) = 1.24e-3+2.678e-3/(1+exp(62.4*(Vm(i,j)+0.050)));
       m(i,j) = m(i,j-1) + dt*(SS_m(i,j)-m(i,j-1))/Tao_m(i,j);
       h(i,j) = h(i,j-1) + dt*(SS_h(i,j)-h(i,j-1))/Tao_h(i,j);
       n(i,j) = n(i,j-1) + dt*(SS_n(i,j)-n(i,j-1))/Tao_n(i,j);
       a(i,j) = a(i,j-1) + dt*(SS_a(i,j)-a(i,j-1))/Tao_a(i,j);
       b(i,j) = b(i,j-1) + dt*(SS_b(i,j)-b(i,j-1))/Tao_b(i,j);
   end
   if(Ib(i) == 900e-12)
       figure(1)
       subplot(2,1,1)
       plot(t,Iapp(i,:)*1e12,'k')
       xlabel("Time (s)")
       ylabel("Applied current (pA)")
       subplot(2,1,2)
       plot(t,Vm(i,:)*1000,'k');
       xlabel("Time (s)")
       ylabel("Membrane potential (mV)")
       figure(2)
       plot(t,m(i,:),"red")
       hold on
       plot(t,h(i,:),"blue")
       plot(t,n(i,:),"cyan")
       plot(t,a(i,:),"green")
       plot(t,b(i,:),"yellow")
       xlabel("Time (s)")
       ylabel("Gating variables")
   end
```

```
   firerates(i) = firecounts(i)/1; %%the current only applied for 1s. I could just not write
firerates, but whatever~ ;)
end
figure(3)
plot(Ib*1e12,firerates)
xlabel("Applied current (pA)")
ylabel("Fire Rate (Hz)")
```

code_9_Thalamic_Rebound_Model.m

```
function [Vm,t,spike,min_interval] = PIR(Iapp,dt,tmax)
   spike = 0;
   min_interval = 1e5;
   state = 0;
   spnow = 0;
   splast = 0;
   Gl = 10e-9;
   GMNa = 3.6e-6;
   GMK = 1.6e-6;
   GMT = 0.22e-6;
   ENa = 55e-3;
   EK = -90e-3;
   ECa = 120e-3;
   El = -70e-3;
   Cm = 100e-12;
   t = dt:dt:tmax;
   Vm = zeros(size(t));
   m = zeros(size(t));
   h = zeros(size(t));
   n = zeros(size(t));
   mt = zeros(size(t));
   ht = zeros(size(t));
   alpha_m = zeros(size(t));
   alpha_h = zeros(size(t));
   alpha_n = zeros(size(t));
   beta_m = zeros(size(t));
   beta_h = zeros(size(t));
   beta_n = zeros(size(t));
   SS_m = zeros(size(t));
   Tao_m = zeros(size(t));
   SS_h = zeros(size(t));
   Tao_h = zeros(size(t));
   SS_n = zeros(size(t));
   Tao_n = zeros(size(t));
   SS_mt = zeros(size(t));
   Tao_mt = zeros(size(t));
   SS_ht = zeros(size(t));
   Tao_ht = zeros(size(t));
   Vm(1) = El;
   m(1) = 0;
   h(1) = 0;
   n(1) = 0;
   mt(1) = 0;
   ht(1) = 0;
   for i=2:length(t)
       Vm(i) =
Vm(i-1)+dt*(Gl*(El-Vm(i-1))+GMNa*m(i-1)^3*h(i-1)*(ENa-Vm(i-1))+GMK*n(i-1)^4*(EK-Vm(i-1))+GMT*m
t(i-1)^2*ht(i-1)*(ECa-Vm(i-1))+Iapp(i-1))/Cm;
           if(Vm(i)>0&&state == 0)
               state = 1;
               spike = spike+1;
               splast = spnow;
               spnow = i;
               min_interval = min(min_interval,(spnow-splast));
           end
           if(Vm(i)<0&&state == 1)
               state = 0;
           end
       if(Vm(i)==-0.035)
           alpha_m(i) = 1000;
       else
```

```
            alpha_m(i) = 1e5*(Vm(i)+0.035)/(1-exp(-100*(Vm(i)+0.035)));
        end
        beta_m(i) = 4000*exp((-Vm(i)-0.06)/0.018);
        alpha_h(i) = 350*exp(-50*(Vm(i)+0.058));
        beta_h(i) = 5000/(1+exp(-100*(Vm(i)+0.028)));
        if(Vm(i)==-0.034)
            alpha_n(i) = 500;
        else
            alpha_n(i) = 5e4*(Vm(i)+0.034)/(1-exp(-100*(Vm(i)+0.034)));
        end
        beta_n(i) = 625*exp(-12.5*(Vm(i)+0.044));
        SS_m(i) = alpha_m(i)/(alpha_m(i)+beta_m(i));
        SS_h(i) = alpha_h(i)/(alpha_h(i)+beta_h(i));
        SS_n(i) = alpha_n(i)/(alpha_n(i)+beta_n(i));
        Tao_m(i) = 0;
        Tao_h(i) = 1/(alpha_h(i)+beta_h(i));
        Tao_n(i) = 1/(alpha_n(i)+beta_n(i));
        SS_mt(i) = 1/(1+exp((-Vm(i)-0.052)/0.0074));
        Tao_mt(i) = 0;
        SS_ht(i) = 1/(1+exp(500*(Vm(i)+0.076)));
        if(Vm(i)<-0.080)
            Tao_ht(i) = 0.001*exp(15*(Vm(i)+0.467));
        else
            Tao_ht(i) = 0.028+0.001*exp((-Vm(i)-0.022)/0.0105);
        end
        m(i) = SS_m(i);
        h(i) = h(i-1) + dt*(SS_h(i)-h(i-1))/Tao_h(i);
        n(i) = n(i-1) + dt*(SS_n(i)-n(i-1))/Tao_n(i);
        mt(i) = SS_mt(i);
        ht(i) = ht(i-1) + dt*(SS_ht(i)-ht(i-1))/Tao_ht(i);
    end
end
dt = 0.01e-3;
tmax = 750e-3;
t = dt:dt:tmax;
Ibmin = -200e-12;
Ibmax = 200e-12;
dI = 10e-12;
Ibase = Ibmin:dI:Ibmax;
Iemin = 0;
Iemax = 100e-12;
Ie = Iemin:dI:Iemax;
spikes = zeros(length(Ibase),length(Ie));
min_ISI = zeros(length(Ibase),length(Ie));
for i = 1:length(Ibase)
    for j= 1:length(Ie)
        Iapp = Ibase(i)*ones(size(t));
        Iapp(round(250e-3/dt):round(500e-3/dt)) = Ie(j);
        [Vm,t,spikes(i,j),min_ISI(i,j)] = PIR(Iapp, dt, tmax);
        if(i==11&&j==1)
            figure(2)
            subplot(2,1,1)
            plot(t,Iapp)
            ylabel("Applied Current (A)")
            subplot(2,1,2)
            plot(t,Vm)
            xlabel("Time (s)")
            ylabel("Vm (V)")
        end
    end
end
figure(1)
subplot(2,1,1)
imagesc(spikes);
title("spikes")
subplot(2,1,2);
imagesc(min_ISI);
title("Interspike Intervals")
```

code_10_Pinsky_Renzel_Model.m:

```
dt = 2e-6;
tmax = 2;
t = dt:dt:tmax;
As = 1/3;
Ad = 1-As;
Gls = As*5e-9;
Gld = Ad*5e-9;
GMNa = As*3e-6;
GMK = As*2e-6;
GMCa = Ad*2e-6;
GMKCa = Ad*2.5e-6;
GMKAHP = Ad*40e-9;
Glink = 50e-9;
ENa = 60e-3;
ECa = 80e-3;
EK = -75e-3;
El = -60e-3;
Cs = As*100e-12;
Cd = Ad*100e-12;
TaoCa = 50e-3;
k = 5e6/Ad;
ISapp = zeros(size(t));
IDapp = zeros(size(t));
m = zeros(size(t));
h = zeros(size(t));
n = zeros(size(t));
mca = zeros(size(t));
mkca = zeros(size(t));
mkahp = zeros(size(t));
Vs = zeros(size(t));
Vd = zeros(size(t));
X = zeros(size(t));
Ca = zeros(size(t));
INa = zeros(size(t));
IK = zeros(size(t));
ICa = zeros(size(t));
IKCa = zeros(size(t));
IGMKAHP = zeros(size(t));
Ilink = zeros(size(t));
IlS = zeros(size(t));
IlD = zeros(size(t));
IS = zeros(size(t));
ID = zeros(size(t));
Vs(1) = El;
Vd(1) = -0.05;
n(1) = 0.4;
h(1) = 0.5;
mkahp(1) = 0.2;
mac(1) = 0.02;
mkca(1) = 0.02;
Ca(1) = 1e-4;
for i = 1:length(t)-1

alpha_m=(Vs(i)==-0.0469)*320e3/250+(Vs(i)~=-0.0469)*320e3*(Vs(i)+0.0469)/(1-exp(-250*(Vs(i)+0.
0469)));

beta_m=(Vs(i)==-0.0199)*280e3/200+(Vs(i)~=-0.0199)*280e3*(Vs(i)+0.0199)/(exp(200*(Vs(i)+0.0199
))-1);
   alpha_h=128*exp(-(Vs(i)+0.043)/0.018);
   beta_h=4000/(1+exp(-200*(Vs(i)+0.020)));

alpha_n=(Vs(i)==-0.0249)*16e3/200+(Vs(i)~=-0.0249)*16e3*(Vs(i)+0.0249)/(1-exp(-200*(Vs(i)+0.02
49)));
   beta_n=250*exp(-25*(Vs(i)+0.040));
   alpha_mca=1600/(1+exp(-72*(Vd(i)-0.005)));

beta_mca=(Vd(i)==-0.0089)*2e4/200+(Vd(i)~=-0.0089)*2e4*(Vd(i)+0.0089)/(exp(200*(Vd(i)+0.0089))
-1);
```

```
alpha_mkca=(Vd(i)>-0.010)*2000*exp((-Vd(i)-0.0535)/0.027)+(Vd(i)<=-0.010)*exp((Vd(i)+0.050)/0.
011-(Vd(i)+0.0535)/0.027)/0.018975;
   beta_mkca=(Vd(i)>-0.010)*0+(Vd(i)<=-0.010)*(2000*exp((-Vd(i)-0.0535)/0.027)-alpha_mkca);
   alpha_mkahp=min(20,20000*Ca(i));
   beta_mkahp=4;

   SS_m = alpha_m/(alpha_m+beta_m);
   Tao_m = 1/(alpha_m+beta_m);
   SS_h = alpha_h/(alpha_h+beta_h);
   Tao_h = 1/(alpha_h+beta_h);
   SS_n = alpha_n/(alpha_n+beta_n);
   Tao_n = 1/(alpha_n+beta_n);
   SS_mca = alpha_mca/(alpha_mca+beta_mca);
   Tao_mca = 1/(alpha_mca+beta_mca);
   SS_mkca = alpha_mkca/(alpha_mkca+beta_mkca);
   Tao_mkca = 1/(alpha_mkca+beta_mkca);
   SS_mkahp = alpha_mkahp/(alpha_mkahp+beta_mkahp);
   Tao_mkahp = 1/(alpha_mkahp+beta_mkahp);
   X(i)=min(4000*Ca(i),1);
   m(i+1) = m(i)+dt*(SS_m-m(i))/Tao_m;
   h(i+1) = h(i)+dt*(SS_h-h(i))/Tao_h;
   n(i+1) = n(i)+dt*(SS_n-n(i))/Tao_n;
   mca(i+1) = mca(i)+dt*(SS_mca-mca(i))/Tao_mca;
   mkca(i+1) = mkca(i)+dt*(SS_mkca-mkca(i))/Tao_mkca;
   mkahp(i+1) = mkahp(i)+dt*(SS_mkahp-mkahp(i))/Tao_mkahp;
   GMNaN = GMNa*m(i)^2*h(i);
   GMKN = GMK*n(i)^2;
   GMCaN = GMCa*mca(i)^2;
   GMKCaN = GMKCa*mkca(i)*X(i);
   GMKAHPN = GMKAHP*mkahp(i);
   INa(i) = GMNaN*(ENa-Vs(i));
   IK(i) = GMKN*(EK-Vs(i));
   ICa(i) = GMCaN*(ECa-Vd(i));
   IKCa(i) = GMKCaN*(EK-Vd(i));
   IGMKAHP(i) = GMKAHP*(EK-Vd(i));
   Ilink(i) = Glink*(Vd(i)-Vs(i));
   IlS(i) = Gls*(El-Vs(i));
   IlD(i) = Gld*(El-Vd(i));
   IS(i) = IlS(i)+INa(i)+IK(i)+Ilink(i)+ISapp(i);
   ID(i) = IlD(i)+ICa(i)+IKCa(i)+IGMKAHP(i)-Ilink(i)+IDapp(i);
   GStot = Gls+GMNaN+GMKN+Glink;
   GDtot = Gld+GMCaN+GMKCaN+GMKAHPN+Glink;
   SS_Vs = (Gls*El+GMNaN*ENa+GMKN*EK+Glink*Vd(i)+ISapp(i))/GStot;
   SS_Vd = (Gld*El+GMCaN*ECa+GMKCaN*EK+GMKAHPN*EK+Glink*Vs(i)+IDapp(i))/GDtot;
   SS_Ca = TaoCa*k*ICa(i);
   Vs(i+1) = SS_Vs-(SS_Vs-Vs(i))*exp(-dt*GStot/Cs);
   Vd(i+1) = SS_Vd-(SS_Vd-Vd(i))*exp(-dt*GDtot/Cd);
   Ca(i+1) = SS_Ca-(SS_Ca-Ca(i))*exp(-dt/TaoCa);
end
figure(1)
subplot(2,1,1)
plot(t,Vd,'k');
ylabel("Vd (V)")
subplot(2,1,2)
plot(t,Vs,'k')
ylabel("Vs (V)")
xlabel("Time (s)")
figure(2)
subplot(2,1,1)
plot(t,Vd,'k');
ylabel("Vd (V)")
axis([0.3 0.4 -0.08 0.02]);
subplot(2,1,2)
plot(t,Vs,'k')
ylabel("Vs (V)")
xlabel("Time (s)")
axis([0.3 0.4 -0.08 0.02]);
```

code_11_Synaptic_Dynamics.m:

```
dt = 0.1e-3;
tmax = 4;
t = dt:dt:tmax;
r = zeros(size(t));
r(1:round(1/dt)) = 20;
r(round(1/dt)+1:round(2/dt)) = 100;
r(round(2/dt)+1:round(3/dt)) = 10;
r(round(3/dt)+1:floor(tmax/dt)) = 50;
spikes = zeros(size(t));
for i = 1:length(t)
   spikes(i) = rand()<dt*r(i);
end
Gsyn = zeros(size(t));
Gsyn(1) = 0;
dGsyn = 1e-9;
TaoGsyn = 100e-3;
for i = 2:length(t)
   Gsyn(i) = Gsyn(i-1) + dt*(-Gsyn(i-1)/TaoGsyn);
   if(spikes(i))
       Gsyn(i) = Gsyn(i) +dGsyn;
   end
end
p0 = 0.5;
D = zeros(size(t));
TaoD = 0.25;
D(1) = 1;
for i = 2:length(t)
   D(i) = D(i-1) +dt*(1-D(i-1))/TaoD;
   if(spikes(i))
       D(i) = D(i)*(1-p0);
   end
end
Gsyn_2 = zeros(size(t));
TaoGsyn_2 = 100e-3;
dGsyn_2max = 5e-9;
p0G = 0.5;
for i = 2:length(t)
   Gsyn_2(i) = Gsyn_2(i-1) + dt*(-Gsyn_2(i-1)/TaoGsyn_2);
   if(spikes(i))
       Gsyn_2(i) = Gsyn_2(i) +dGsyn_2max*p0G*D(i);
   end
end
p0F = 0.2;
F = zeros(size(t));
F(1) = 1;
TaoF = 0.25;
ffac = 0.25;
Fmax = 1/p0F;
for i = 2:length(t)
   F(i) = F(i-1) + dt*(1-F(i-1))/TaoF;
   if(spikes(i))
       F(i) = F(i) +ffac*(Fmax-F(i));
   end
end
D2 = zeros(size(t));
TaoD = 0.25;
D2(1) = 1;
for i = 2:length(t)
   D2(i) = D2(i-1) +dt*(1-D2(i-1))/TaoD;
   if(spikes(i))
       D2(i) = D2(i)*(1-p0F*F(i));
   end
end
Gsyn_3 = zeros(size(t));
TaoGsyn_3 = 100e-3;
dGsyn_3max = 4e-9;
p0G2 = 0.2;
for i = 2:length(t)
   Gsyn_3(i) = Gsyn_3(i-1) + dt*(-Gsyn_3(i-1)/TaoGsyn_3);
   if(spikes(i))
       Gsyn_3(i) = Gsyn_3(i) +dGsyn_3max*p0G2*D(i)*F(i);
```

```
   end
end
clf
figure(1);
subplot(3,2,1);
plot(t,Gsyn)
xlabel("Gsyn")
subplot(3,2,2)
plot(t,D);
xlabel("D")
subplot(3,2,3)
plot(t,Gsyn_2);
xlabel("Gsyn_2");
subplot(3,2,4)
plot(t,F);
xlabel("F")
subplot(3,2,6)
plot(t,D2);
xlabel("D2")
subplot(3,2,5)
plot(t,Gsyn_3);
xlabel("Gsyn_3")
```

code_12_Synaptic_Dynamics_With_A_Postsynaptic_Neuron.m:

```
dt = 0.1e-3;
tmax = 4;
t = dt:dt:tmax;
r = zeros(size(t));
r(1:round(1/dt)) = 20;
r(round(1/dt)+1:round(2/dt)) = 100;
r(round(2/dt)+1:round(3/dt)) = 10;
r(round(3/dt)+1:floor(tmax/dt)) = 50;
spikes = zeros(size(t));
for i = 1:length(t)
   spikes(i) = rand()<dt*r(i);
end
Gsyn = zeros(size(t));
Gsyn(1) = 0;
dGsyn = 1e-9;
TaoGsyn = 100e-3;
for i = 2:length(t)
   Gsyn(i) = Gsyn(i-1) + dt*(-Gsyn(i-1)/TaoGsyn);
   if(spikes(i))
       Gsyn(i) = Gsyn(i) +dGsyn;
   end
end
%e.
p0 = 0.5;
D = zeros(size(t));
TaoD = 0.25;
D(1) = 1;
for i = 2:length(t)
   D(i) = D(i-1) +dt*(1-D(i-1))/TaoD;
   if(spikes(i))
       D(i) = D(i)*(1-p0);
   end
end
%f.
Gsyn_2 = zeros(size(t));
TaoGsyn_2 = 100e-3;
dGsyn_2max = 5e-9;
p0G = 0.5;
for i = 2:length(t)
   Gsyn_2(i) = Gsyn_2(i-1) + dt*(-Gsyn_2(i-1)/TaoGsyn_2);
   if(spikes(i))
       Gsyn_2(i) = Gsyn_2(i) +dGsyn_2max*p0G*D(i);
   end
end
%g
p0F = 0.2;
```

```
F = zeros(size(t));
F(1) = 1;
TaoF = 0.25;
ffac = 0.25;
Fmax = 1/p0F;
for i = 2:length(t)
   F(i) = F(i-1) + dt*(1-F(i-1))/TaoF;
   if(spikes(i))
       F(i) = F(i) +ffac*(Fmax-F(i));
   end
end
%h
D2 = zeros(size(t));
TaoD = 0.25;
D2(1) = 1;
for i = 2:length(t)
   D2(i) = D2(i-1) +dt*(1-D2(i-1))/TaoD;
   if(spikes(i))
       D2(i) = D2(i)*(1-p0F*F(i));
   end
end
%i
Gsyn_3 = zeros(size(t));
TaoGsyn_3 = 100e-3;
dGsyn_3max = 4e-9;
p0G2 = 0.2;
for i = 2:length(t)
   Gsyn_3(i) = Gsyn_3(i-1) + dt*(-Gsyn_3(i-1)/TaoGsyn_3);
   if(spikes(i))
       Gsyn_3(i) = Gsyn_3(i) +dGsyn_3max*p0G2*D(i)*F(i);
   end
end
El = -65e-3;
Esyn = 0;
Gl = 2e-9;
Cm = 20e-12;
Vm = zeros(3,length(t));
Vth = -50e-3;
Vreset = -80e-3;
for i = 1:3
   switch i
       case 1
           Gs = Gsyn;
       case 2
           Gs = Gsyn_2;
       case 3
           Gs = Gsyn_3;
   end
   Vm(i,1) = El;
   for j = 2:length(t)
       Vm(i,j) = Vm(i,j-1) + dt*(Gl*(El-Vm(i,j-1))+Gs(j)*(Esyn-Vm(i,j-1)))/Cm;
       if(Vm(i,j)>Vth)
           Vm(i,j) = Vreset;
       end
   end
   figure(1);
   subplot(3,1,i);
   plot(t,Vm(i,:));
end
```

code_13_Neuron_Connectivity.m:

```
Ncell = 300;
Ne = zeros(1,Ncell);
for i = 1:Ncell
   Ne(i) = ceil(10*rand());
end
C = zeros(Ncell);
for i = 1:Ncell
   for j = 1:Ncell
       C(i,j) = rand() < 0.1*(~Ne(i)==Ne(j)) + 0.5*(Ne(i)==Ne(j));
```

```
   end
end
figure(1)
imagesc(C);
Corr_C1 = corr(C);
Corr_C2 = corr(C');
Corr_Ctot = (Corr_C2+Corr_C1)/2;
figure(2)
histogram(Corr_Ctot)
seqn = sort(Corr_Ctot(:));
th = seqn(round(Ncell^2*0.9));
%k = 0.15/(1+(exp(-Ngroup)*35000))+0.80;
%th = seqn(round(Ncell^2*k));
%Those two lines can be added to optimize the threshold.
Cell_Group = zeros(1,Ncell);
Group_count = 0;
for i = 1:length(Cell_Group)
   if(Cell_Group(i)==0)
       Group_count = Group_count+1;
       for j = i:length(Cell_Group)
           if(Corr_Ctot(i,j) >= th && Cell_Group(j) == 0)
               Cell_Group(j) = Group_count;
           end
       end
   end
end
[B,new_order] = sort(Cell_Group);
C2 = C(new_order,new_order);
figure(3)
imagesc(C2);
```

code_14_Bistability.m:

```
C = 1e-9;
R = 10e6;
E = -70e-3;
Vth = -54e-3;
Vr = -80e-3;
Er12 = -70e-3;
Er21 = -70e-3;
G12 = 1e-6;
G21 = 1e-6;
Taosyn = 10e-3;
Ibase = 2e-9;
Iextra = 3e-9;
pr = 0.2;
tmax = 6;
dt = 10e-6;
t = dt:dt:tmax;
V1 = zeros(size(t));
V2 = zeros(size(t));
s1 = zeros(size(t));
s2 = zeros(size(t));
D1 = zeros(size(t));
D2 = zeros(size(t));
TaoD = 0.2;
V1(1) = E;
V2(1) = E;
Iapp1 = ones(size(t))*Ibase;
Iapp2 = ones(size(t))*Ibase;
Iapp1(1:round(0.1/dt)) = Iapp1(1:round(0.1/dt)) + Iextra;
Iapp2(round(3/dt):round(3.1/dt)) = Iapp2(round(3/dt):round(3.1/dt)) + Iextra;
for i = 2:length(t)
   V1(i) = V1(i-1) + dt*((E-V1(i-1))/R+G21*s2(i-1)*(Er21-V1(i-1))+Iapp1(i-1))/C;
   V2(i) = V2(i-1) + dt*((E-V2(i-1))/R+G12*s1(i-1)*(Er12-V2(i-1))+Iapp2(i-1))/C;
   s1(i) = s1(i-1) - dt*s1(i-1)/Taosyn;
   s2(i) = s2(i-1) - dt*s2(i-1)/Taosyn;
   D1(i) = D1(i-1) + dt*(1-D1(i-1))/TaoD;
   D2(i) = D2(i-1) + dt*(1-D2(i-1))/TaoD;
   if(V1(i)>Vth)
       V1(i) = Vr;
```

```
        s1(i) = s1(i)+pr*D1(i)*(1-s1(i));
        D1(i) = D1(i)*(1-pr);
    end
    if(V2(i)>Vth)
        V2(i) = Vr;
        s2(i) = s2(i)+pr*D2(i)*(1-s2(i));
        D2(i) = D2(i)*(1-pr);
    end
end
clf
figure(1)
subplot(2,1,1)
plot(t,V1,'r');
hold on;
plot(t,V2,'b');
xlabel("Time (s)")
ylabel("Membrane potential (V)")
subplot(2,1,2);
plot(t,V1,'r');
axis([2.75 3.25 -0.08 -0.05]);
hold on;
plot(t,V2,'b');
axis([2 3 -0.08 -0.05]);
xlabel("Time (s)")
ylabel("Membrane potential (V)");
```

code_15_Oscillations.m:

```
C = 1e-9;
R = 10e6;
E = -70e-3;
Vth = -54e-3;
Vr = -80e-3;
Er12 = -70e-3;
Er21 = -70e-3;
G12 = 1e-6;
G21 = 1e-6;
Taosyn = 10e-3;
Ibase = 2e-9;
Iextra = 3e-9;
pr = 0.2;
tmax = 6;
dt = 10e-6;
t = dt:dt:tmax;
V1 = zeros(size(t));
V2 = zeros(size(t));
s1 = zeros(size(t));
s2 = zeros(size(t));
D1 = zeros(size(t));
D2 = zeros(size(t));
TaoD = 0.2;
V1(1) = E;
V2(1) = E;
Iapp1 = ones(size(t))*Ibase;
Iapp2 = ones(size(t))*Ibase;
Iapp1(1:round(0.1/dt)) = Iapp1(1:round(0.1/dt)) + Iextra;
Iapp2(round(3/dt):round(3.1/dt)) = Iapp2(round(3/dt):round(3.1/dt)) + Iextra;
for i = 2:length(t)
    V1(i) = V1(i-1) + dt*((E-V1(i-1))/R+G21*s2(i-1)*(Er21-V1(i-1))+Iapp1(i-1))/C;
    V2(i) = V2(i-1) + dt*((E-V2(i-1))/R+G12*s1(i-1)*(Er12-V2(i-1))+Iapp2(i-1))/C;
    s1(i) = s1(i-1) - dt*s1(i-1)/Taosyn;
    s2(i) = s2(i-1) - dt*s2(i-1)/Taosyn;
    D1(i) = D1(i-1) + dt*(1-D1(i-1))/TaoD;
    D2(i) = D2(i-1) + dt*(1-D2(i-1))/TaoD;
    if(V1(i)>Vth)
        V1(i) = Vr;
        s1(i) = s1(i)+pr*D1(i)*(1-s1(i));
        D1(i) = D1(i)*(1-pr);
    end
    if(V2(i)>Vth)
        V2(i) = Vr;
```

```
        s2(i) = s2(i)+pr*D2(i)*(1-s2(i));
        D2(i) = D2(i)*(1-pr);
    end
end
clf
figure(1)
subplot(2,1,1)
plot(t,V1,'r');
hold on;
plot(t,V2,'b');
xlabel("Time (s)")
ylabel("Membrane potential (V)")
subplot(2,1,2);
plot(t,V1,'r');
axis([2.75 3.25 -0.08 -0.05]);
hold on;
plot(t,V2,'b');
axis([2 3 -0.08 -0.05]);
xlabel("Time (s)")
ylabel("Membrane potential (V)");
```

code_16_Fire_Rate_Model:

```
clear
x = 1.2;
r0 = -0.1;
rmax = 100;
sigma = 0.5;
Taor = 10e-3;
alpha0 = 0.25;
WEE = 35;
pr = 1;
Taos = 2e-3;
ds = 1e-4;
S = ds:ds:1/WEE;
f = zeros(size(S));
dr = 1e-3;
R = dr:dr:rmax;
syD = zeros(size(R));
TaoD = 125e-3;
for i = 1:length(S)
    f(i) = r0+rmax*(WEE^x*S(i)^x/(WEE^x*S(i)^x+sigma^x));
end
for i = 1:length(R)
    syD(i) =
alpha0*(1/(1+pr*R(i)*TaoD))*pr*R(i)*Taos/(1+alpha0*(1/(1+pr*R(i)*TaoD))*pr*R(i)*Taos);
end
clf
figure(1)
plot(S,f,'r');
hold on
plot(syD,R,'b');
xlabel("s")
ylabel("rate (Hz)")
tmax = 20;
dt = 1e-3;
t = dt:dt:tmax;
s = zeros(size(t));
r = zeros(size(t));
sb = 0.05;
sin = zeros(size(t));
sin(10/dt:10.05/dt)=sb;
D = zeros(size(t));
r(1) = 0;
s(1) = 0;
fs = zeros(size(t));
for i = 2:length(t)
    if(WEE*s(i-1)+sin(i-1)>0)
        fs(i-1) = r0+rmax*((WEE*s(i-1)+sin(i-1))^x/((WEE*s(i-1)+sin(i-1))^x+sigma^x));
    else
        fs(i-1) = r0;
```

```
    end
    r(i) = r(i-1)+dt*(-r(i-1)+fs(i-1))/Taor;
    D(i) = D(i-1)+dt*(((1-D(i-1))/TaoD)-pr*D(i-1)*r(i-1));
    s(i) = s(i-1)+dt*(-s(i-1)/Taos+alpha0*D(i-1)*pr*r(i-1)*(1-s(i-1)));
end
figure(2)
plot(t,r)
xlabel("time (s)");
ylabel("Rate (Hz)");
```

code_17_Decision_Making_Circuit.m:

```
G = 1;
sigma_int = 0.25;
sigma_s = 0.2;
tmax = 10;
dt = 0.0005;
t = dt:dt:tmax;
Ntrail = 200;
r1 = zeros(Ntrail,length(t));
r2 = zeros(Ntrail,length(t));
Ws = 0.975;
Wx = -0.025;
theta = -0.5;
Tao = 10e-3;
r1(:,1) = max(0,theta/(Ws+Wx-1));
r2(:,1) = max(0,theta/(Ws+Wx-1));
d_th = 50;
rmax = 60;
ds = zeros(size(t));
mu_s = zeros(size(t));
ds(0.5/dt:tmax/dt) = 0.1;
mu_s(0.5/dt:tmax/dt) = 1;
noise_interval = 2e-3;
noise_bin = noise_interval/dt;
eta_s1 = zeros(Ntrail,length(t));
eta_s2 = zeros(Ntrail,length(t));
eta_1 = zeros(Ntrail,length(t));
eta_2 = zeros(Ntrail,length(t));
mu_firerate1 = zeros(size(t));
mu_firerate2 = zeros(size(t));
cross_th_t1 = zeros(1,Ntrail);
for j = 1:Ntrail
   for i = 1:length(t)
       eta_1(j,i*noise_bin+1-noise_bin:i*noise_bin) = randn()/sqrt(noise_interval);
       eta_2(j,i*noise_bin+1-noise_bin:i*noise_bin) = randn()/sqrt(noise_interval);
       eta_s1(j,i*noise_bin+1-noise_bin:i*noise_bin) =
randn()/sqrt(noise_interval)*max(1,min(1,mu_s(i)));
       eta_s2(j,i*noise_bin+1-noise_bin:i*noise_bin) =
randn()/sqrt(noise_interval)*max(1,min(1,mu_s(i)));
   end
end
for j = 1:Ntrail
   for i = 2:length(t)
       r1(j,i) = r1(j,i-1) +
dt*(-r1(j,i-1)+Ws*r1(j,i-1)+Wx*r2(j,i-1)+G*(mu_s(i-1)+ds(i-1)/2+sigma_s*eta_s1(j,i-1))+sigma_i
nt*eta_1(j,i-1)-theta)/Tao;
       r2(j,i) = r2(j,i-1) +
dt*(-r2(j,i-1)+Ws*r2(j,i-1)+Wx*r1(j,i-1)+G*(mu_s(i-1)-ds(i-1)/2+sigma_s*eta_s2(j,i-1))+sigma_i
nt*eta_2(j,i-1)-theta)/Tao;
       r1(j,i) = min(r1(j,i),rmax);
       r1(j,i) = max(r1(j,i),0);
       r2(j,i) = min(r2(j,i),rmax);
       r2(j,i) = max(r2(j,i),0);
       if(r1(j,i)>d_th&&cross_th_t1(j)==0)
           cross_th_t1(j) = i;
       end
       mu_firerate1(i) = mu_firerate1(i)+r1(j,i)/Ntrail;
       mu_firerate2(i) = mu_firerate2(i)+r2(j,i)/Ntrail;
   end
```

```
end
clf
figure(1)
plot(t,mu_firerate1,'r');
hold on;
plot(t,mu_firerate2,'b');
figure(2)
histogram(cross_th_t1)
figure(3)
plot(t,r1(1,:),'r')
hold on
plot(t,r2(1,:),'b')
```